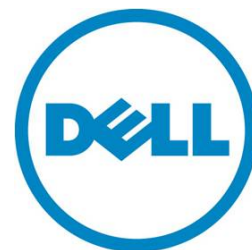

Integrating iDRAC 7 with Microsoft Active Directory

Whitepaper



Author: Jim Slaughter

This document is for informational purposes only and may contain typographical errors and technical inaccuracies. The content is provided as is, without express or implied warranties of any kind.

© 2012 Dell Inc. All rights reserved. Dell and its affiliates cannot be responsible for errors or omissions in typography or photography. Dell, the Dell logo, and PowerEdge are trademarks of Dell Inc. Intel and Xeon are registered trademarks of Intel Corporation in the U.S. and other countries. Microsoft, Windows, and Windows Server are either trademarks or registered trademarks of Microsoft Corporation in the United States and/or other countries. Other trademarks and trade names may be used in this document to refer to either the entities claiming the marks and names or their products. Dell disclaims proprietary interest in the marks and names of others.

January 2012| Rev 1.0

Contents

Benefits of integrating iDRAC7 with Microsoft Active Directory	1
Standard Schema versus Extended Schema	1
This document	1
Confirming the iDRAC7 has an enterprise license installed	1
Dell test environment	2
Building the domain controller	3
Promoting the server to a domain controller and installing DNS	3
Installing and configuring Active Directory Certificate Services	5
Creating iDRAC users and groups	8
Configure iDRAC for use with Active Directory Standard Schema	9
Configure the iDRAC network settings	9
Configure the iDRAC Directory Services settings	11
Test your Standard Schema configuration.....	14
Active Directory login syntax options	16
Authentication examples	17
Example Racadm Command using Active Directory credentials	17
Example WSMAN (WinRM) Command using Active Directory credentials	17
Example SSH login using Active Directory credentials.....	17
iDRAC GUI login examples using Active Directory credentials	18
Configuring the domain controller for use with Active Directory Extended Schema.....	19
Extending the schema	19
Optional - View the changes made to the Active Directory Schema	20
Install the Dell Extension to the Active Directory Users and Computers Snap-In for 32-bit Windows .	21
Install the Dell Extension to the Active Directory Users and Computers Snap-In for 64-bit Windows .	22
Install the Active Directory Users and Computers Snap-In to MMC	22
Privilege and role name changes	23
Active Directory objects	23
Privilege Objects.....	24
iDRAC objects	24
Association objects	25
Configure Active Directory.....	25
Adding users	29
Adding iDRACs	29

Integrating iDRAC7 with Active Directory

Configuring iDRAC for use with Active Directory Extended Schema	29
Test your Extended Schema configuration	31
Creating an Active Directory user with customized iDRAC privileges.....	33
Summary.....	36

Benefits of integrating iDRAC7 with Microsoft Active Directory

Using your integrated Dell Remote Access Controller v7 (iDRAC7) with Microsoft Active Directory simplifies user account and privilege management. It eliminates configuring each individual user and their associated privileges on every iDRAC. Once configured, users provide their Active Directory credentials to authenticate to all iDRACs. You can use these credentials to log into the iDRAC GUI, SSH and Telnet consoles, and for running racadm, and WSMAN commands from the CLI.

Note: You must have an Enterprise License installed on your iDRAC7 to use Active Directory authentication. See [Confirming the iDRAC7 has an Enterprise License Installed](#) for more information.

Standard Schema versus Extended Schema

There are two options when integrating your iDRAC: Standard Schema or Extended Schema, with different advantages and requirements for each.

With either Standard or Extended Schema, existing Active Directory you can easily assign users to groups that have predefined privilege levels for the iDRACs.

Standard Schema has the advantage of not having to extend the Active Directory schema. However, you must enter Active Directory group names and privileges on each iDRAC.

Extended Schema by definition requires an extension to the Active Directory schema, which is an irreversible process. However, this provides the additional benefit of only having to configure the Active Directory group names and privileges once for all iDRACs on the Domain Controller.

This document

Integrating iDRAC with Active Directory can be complex, and this document simplifies the process with step-by-step instructions. There are multiple ways to achieve the same results and steps vary with different operating systems and in different network environments.

This document covers a standard schema setup, then adds extended schema. This lets you get hands-on experience with each option and determine which method is best for you. Once set up, you can switch from either schema method using the same Domain Controller with minimal configuration changes.

It is strongly recommended that you do these steps in a test environment first. Then you can determine the level of integration that works best for you, along with the steps required to implement Active Directory in your environment.

This document assumes you have some experience working in Active Directory on a Domain Controller and are familiar with IP addressing, DNS, and DHCP.

For additional information on integrating iDRAC7 with Active Directory, see the *iDRAC7 User's Guide* on www.support.dell.com.

Confirming the iDRAC7 has an enterprise license installed

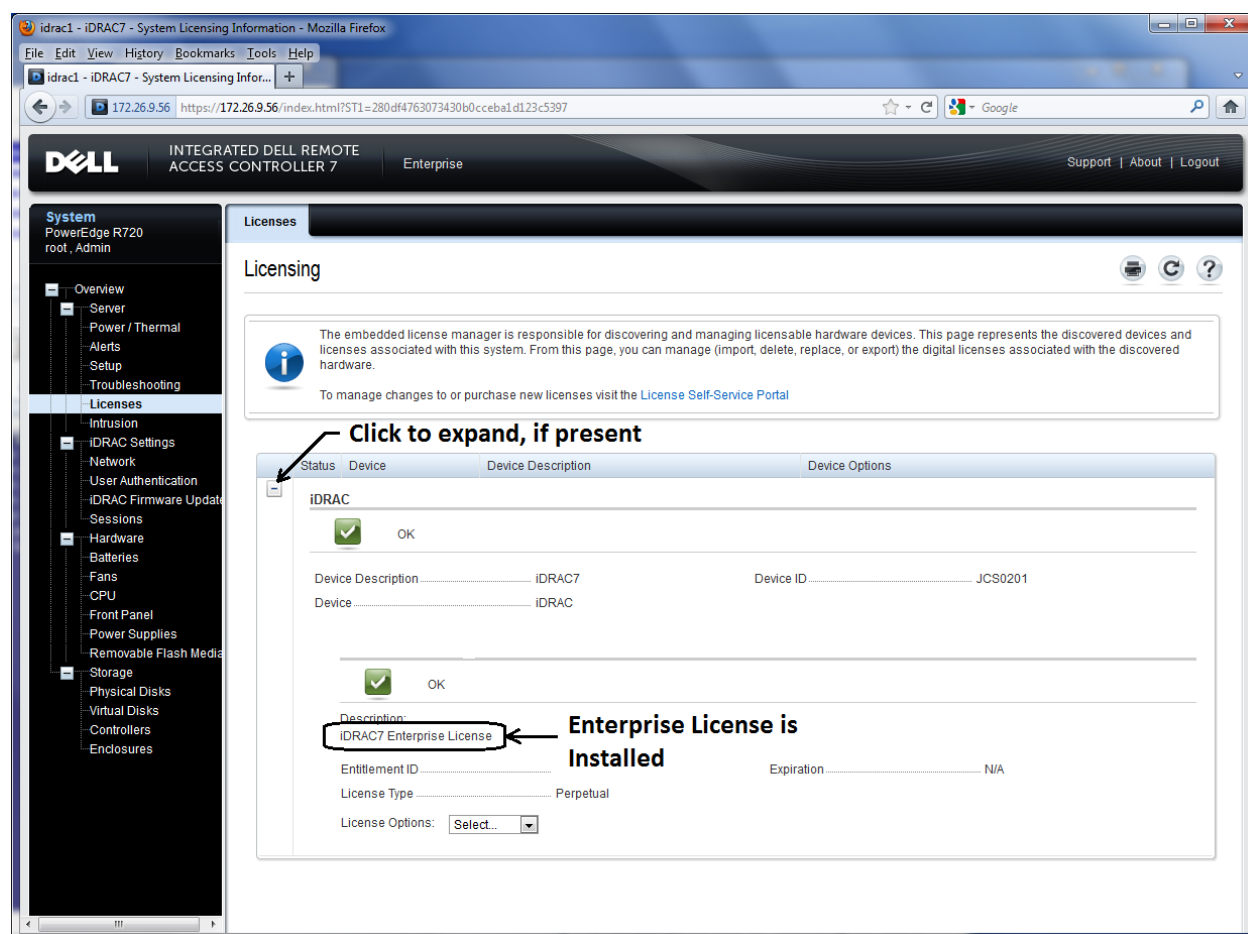
You must have an Enterprise License installed on your iDRAC7 to use Active Directory authentication.

To check the license you have:

Integrating iDRAC7 with Active Directory

1. Browse to **https://<idrac_ip_address>** and log into the iDRAC GUI of your system as an administrative user (default username is root, password is calvin.)
2. Go to the **Overview > Server > Licenses** page.
3. Expand the "+" in the left column of the license table to view the license (pictured below). If you have no "+" to expand, or if the license displayed is "Basic" or "Express", you cannot use the Active Directory feature. However, you can quickly upgrade to an Enterprise License electronically by using the License Self-Service Portal (linked on the Licensing page) or by contacting your Dell Sales representative.

Figure 1. Viewing the license.



Dell test environment

To help you transfer the steps outlined in this document to your environment, the Dell test environment set up is as follows:

Systems Used in the Dell test environment

- **Domain Controller** - A system running Windows Server 2008 Enterprise 32-bit Service Pack 1.

- **Managed System** - A Dell PowerEdge R720 with iDRAC7 and an Enterprise License installed.
- **Management Station** - A system running Windows 7 and Firefox 7. (Internet Explorer is also supported).

Note: See the Systems Management Tools and Documentation DVD version 7.0.0 or later, or www.support.dell.com, for the full list of supported PowerEdge Systems, Operating Systems, and Browsers.

Additional Information about the Dell test environment

- The Active Directory domain name is test.lab.
- The FQDN of the Domain Controller is ad2.test.lab and it has a static IP address of 172.23.199.28.
- DHCP is running on the network on a different server. It is being used to assign an IP address to the iDRAC. (DHCP is optional).
- The iDRAC has been assigned a dynamic IP address of 172.26.9.56.
- DNS and Certificate Services will be running on the Domain Controller (described in *Building the domain controller*).
- The Dell Systems Management Tools and Documentation DVD, version 7.0.0, is used.

Building the domain controller

All of the steps in this section take place on the server used as the *Domain Controller*.

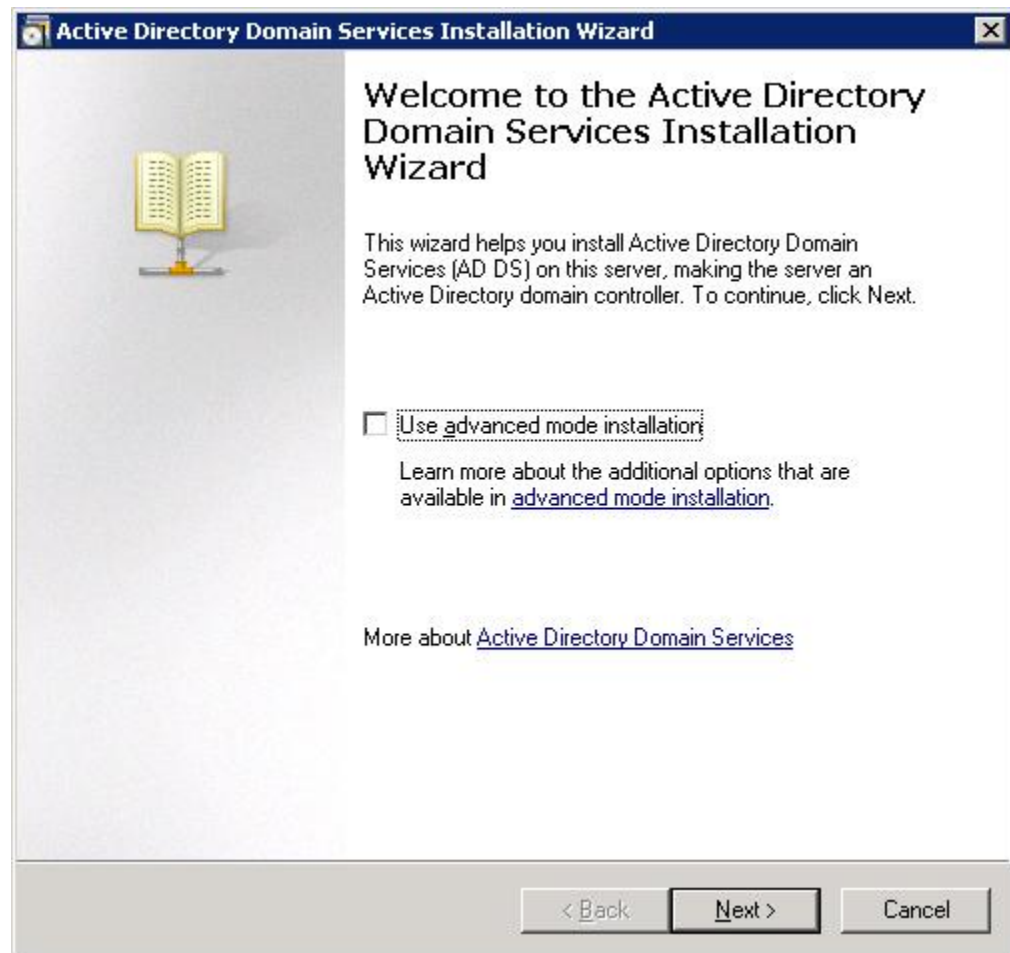
- Install a supported Windows Server operating system, such as Windows Server 2008 Enterprise.
- Make sure the date, time, and time zone on the server are correct. This is critical for Active Directory authentication with iDRAC.
- Configure a static IP address (recommended since it also is the DNS server).
- If desired, change the Windows computer name of the Domain Controller before running the next steps.

Promoting the server to a domain controller and installing DNS

The steps that follow are from Windows Server 2008 Enterprise. Steps for other supported Windows Server operating systems are similar.

1. Promote the server to a Domain Controller. Click **Start > Run > dcpromo**.
2. In the **Active Directory Domain Services Installation Wizard**, click **Next**.

Figure 2. Active Directory Domain Services Installation Wizard.



3. In the Operating System Compatibility page, click **Next**.
 4. Select **Create a new domain in a new forest**, and then click **Next**.
 5. Provide the **FQDN of the forest root domain** (for example, test.lab).
 6. For both Forest & Domain functional levels, choose either **Windows Server 2003** or **Windows Server 2008**, and then click **Next** two times.
- If DNS is not already installed, you are prompted to install it. Accept the default options and install DNS.
7. Accept the default locations for the Database, Log files, and SYSVOL, and then click **Next**.
 8. Assign a **Directory Services Restore Mode Administrator Password**, and then click **Next**.
 9. In the **Summary** page, click **Next**.
 10. Allow the installation to complete and reboot the system when prompted.
 11. Your system is now a Domain Controller running DNS.

Note: If DHCP is not already running on your network, you can optionally install it on the Domain Controller at this time or use static IP addresses on your network.

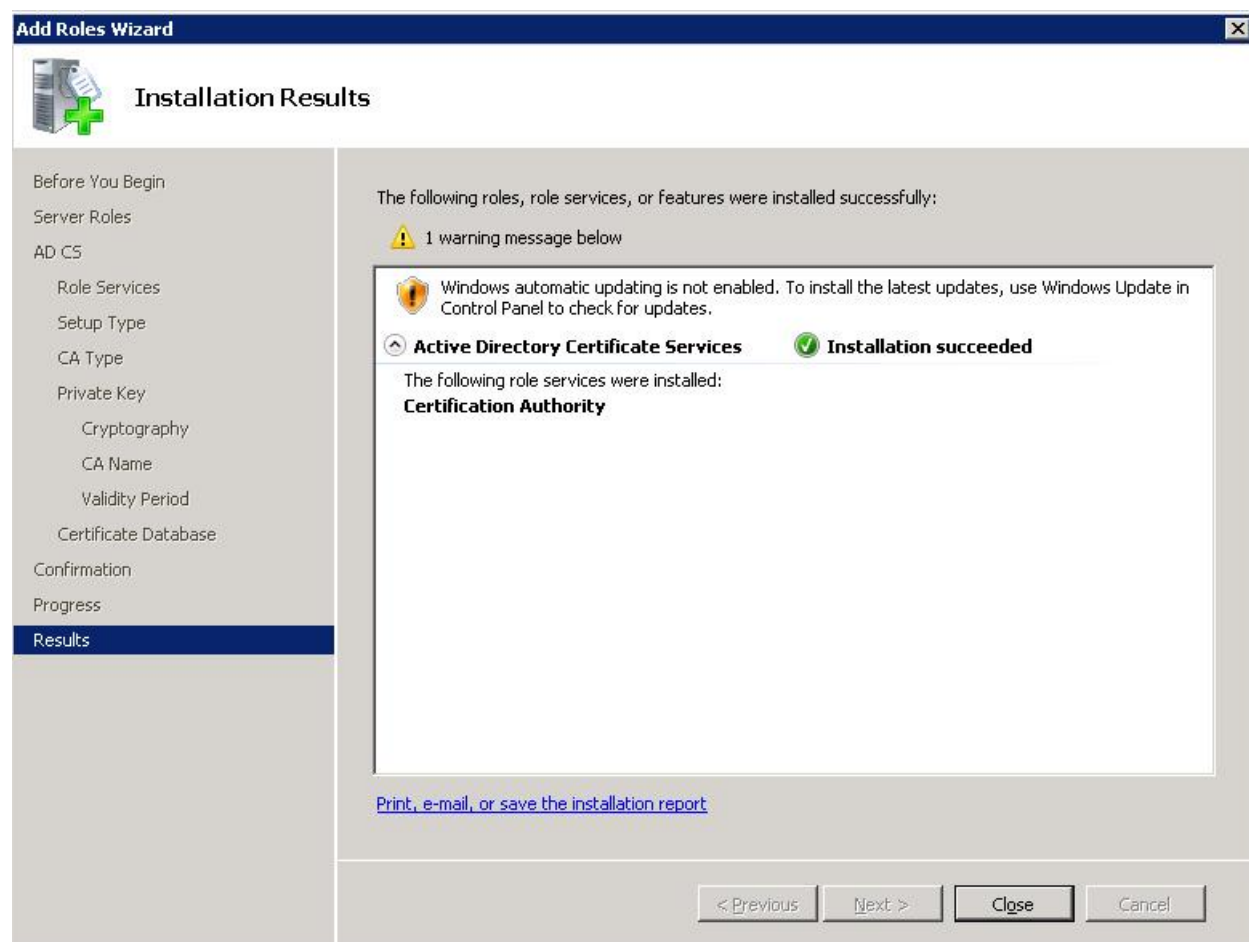
Installing and configuring Active Directory Certificate Services

Installing Certificate Services as an Enterprise Root CA

1. Open **Server Manager**, and then click **Roles > Add Roles**, and then click **Next**.
2. Select **Active Directory Certificate Services**, and then click **Next**.
3. Click **Next**.
4. Make sure **Certification Authority** is checked, then click **Next > Enterprise > Next > Root CA > Next > Create a New Private Key > Next**.
5. Accept the defaults for CSP, key character length, and hash algorithm, and then click **Next**.
6. Accept the default CA name, and then click **Next**.
7. Select the default validity period, and then click **Next**.
8. Select the default database and log locations, and then click **Next**.
9. Click **Install**.

When installation is complete you should get an Installation Succeeded message similar to this:

Figure 3. Installation succeeded message screen.



Adding the certificates snap in to Microsoft Management Console

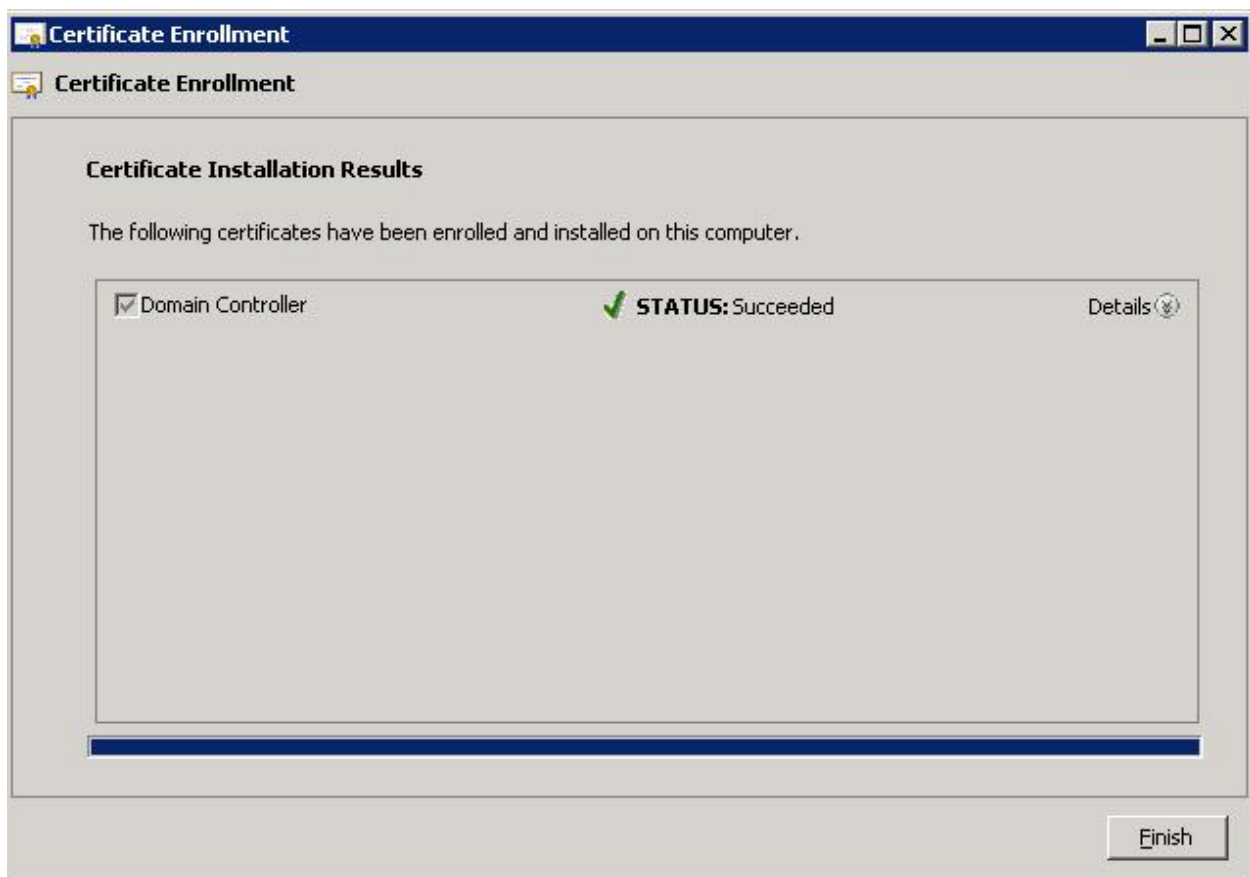
1. Click **Start > Run > MMC > OK**.
2. In the **Console 1** window, click **File > Add/Remove Snap-in > select Certificates > Add > select Computer Account > Next > Local Computer > Finish > OK**.

It is recommended that you save **Console1.msc** to your Desktop at this point. You will use this console for other snap-ins later in this document.

Installing the CA certificate for Client Authentication to the Domain Controller

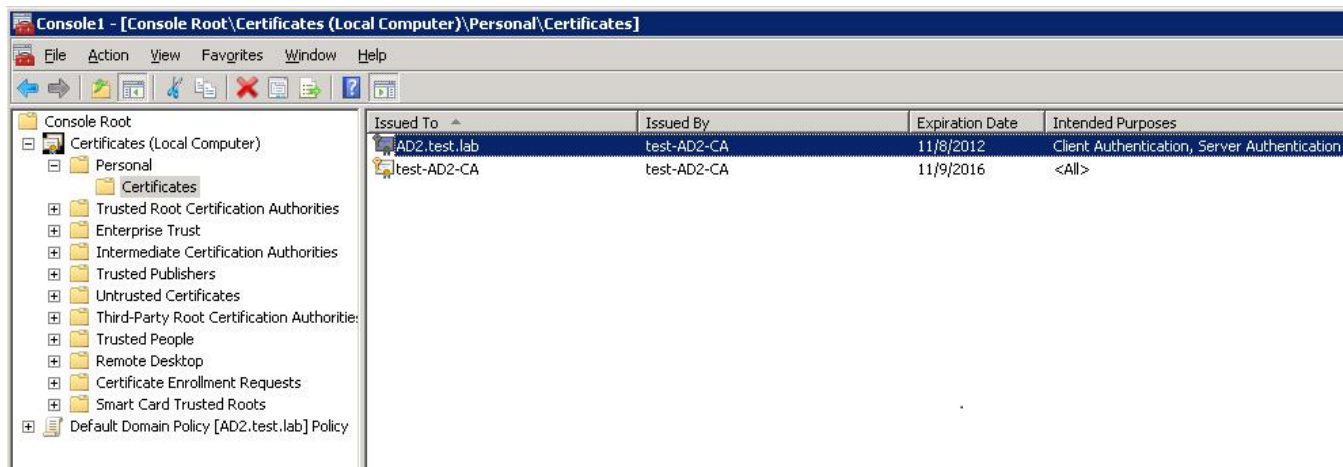
1. Open **Console1**, expand **Certificates**, expand **Personal**, click **Certificates**.
2. Right click **Certificates**, then click **All Tasks > Request New Certificate**.
3. In the **Certificate Enrollment** wizard, click **Next**.
4. Select **Domain Controller**, and click **Enroll > Finish**.
5. You will get the following success message:

Figure 4. Certificate Enrollment success message.



The contents of your certificate folder should now look similar to the following, with the newly created certificate highlighted below.

Figure 5. Certificate folder contents.



Exporting the CA Certificate (You install this certificate on iDRAC later).

1. Locate the **CA** certificate. This is the certificate issued to your CA, (named **test-AD2-CA** in this example).
2. Right click the **CA Certificate** and select **All Tasks > Export**.
3. In the **Certificate Export Wizard** click **Next > No, do not export the private key**, and then click **Next**.
4. Select **Base-64 encoded X.509 (.CER)**, and then click **Next**.
5. Browse to the desired path and give it a file name (For example, **ad2.cer**), and then click **Next**.

Figure 6. Completing the Certificate Export Wizard.



6. Click **Finish**.
7. View the success message and then click **OK**.

Creating iDRAC users and groups

1. In the left pane of **Server Manager**, expand **Roles > Active Directory Domain Services > Active Directory Users and Computers > your domain name (test.lab)**.
2. In the **Users** container, create users that will be provided the three different iDRAC privilege levels. (Right click on **Users** and select **New > User**). For example, create three users and name them:

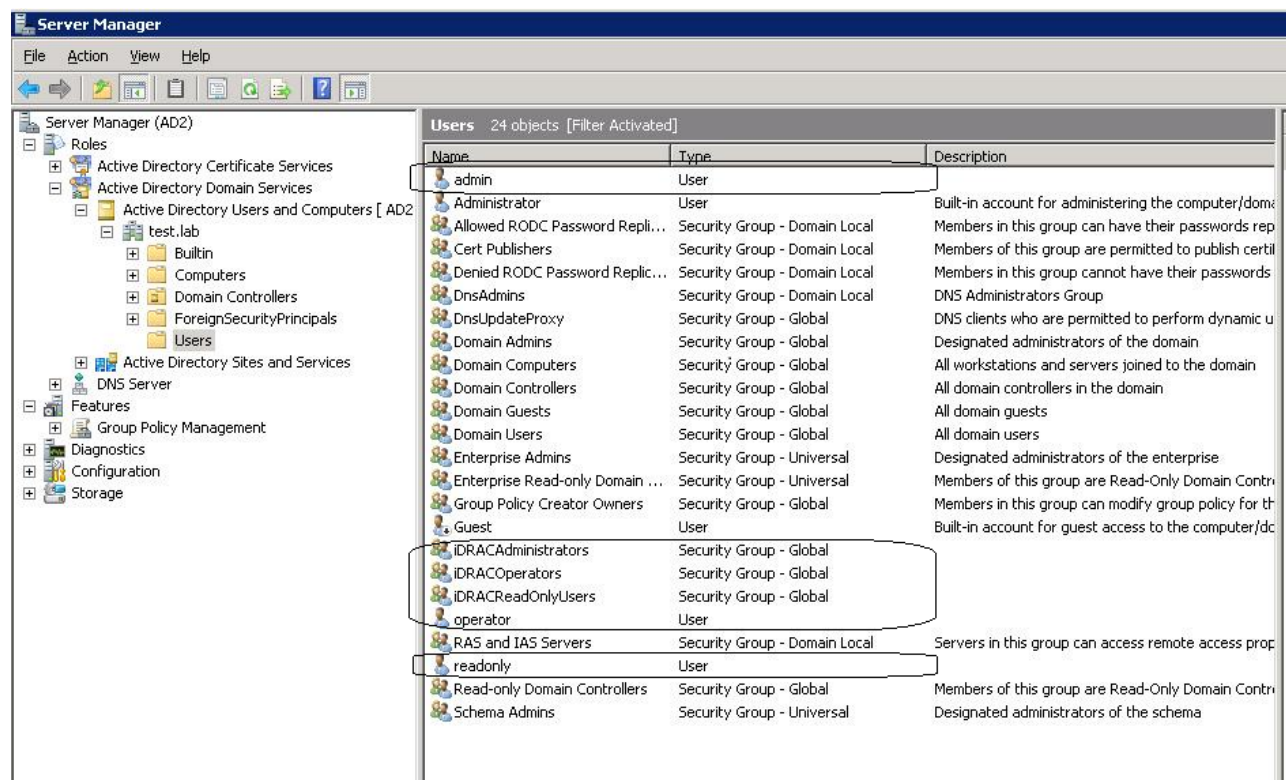
- admin
- operator
- readonly

Note: usernames must be an ASCII string of 1–256 bytes. White space and special characters (such as \, /, or @) cannot be used in the user name.

- Assign each user a password and uncheck **User must change password at next logon** as each user is created.
- In addition, in the **Users** container, create groups based on iDRAC privilege levels that the iDRAC users will belong to (Right click on **Users** and select **New > Group**). Keep the default group type of **Global, Security**). For example, create three groups and name them:
 - o iDRACAdministrators
 - o iDRACOperators
 - o iDRACReadOnlyUsers

When complete, it should look something like this:

Figure 7. iDRAC users and groups



Assigning the users to their corresponding groups

1. Double click on the **admin** user, and then on the **Member Of** tab, click **Add**.
2. Under **Enter the object names to select**, type **iDRAC** (or part of the group name you used) > **Check Names** > select the **iDRACAdministrators** group > **OK** > **OK** > **OK**.
3. Repeat for the **operator** and **readonly** users. (Assign them to **iDRACOperators** and **iDRACReadOnlyUsers** groups respectively.)

Configure iDRAC for use with Active Directory Standard Schema

At the management station, in your Internet Explorer or Firefox web browser, browse to https://<idrac_ip_address> and log into the iDRAC GUI of your system as an administrative user (default username is **root**, password is **calvin**.)

Configure the iDRAC network settings

1. In the iDRAC GUI, go to **iDRAC Settings** > **Network** > **Common Settings**.
 - **Register DRAC on DNS** (unchecked, optional)
 - **DNS DRAC name** (optional), the default is **idrac-<Dell service tag #>**
 - **Auto config domain name (not checked)**. (Only check this if your DHCP server will provide the domain name).

- **Static DNS Domain Name** - specify the FQDN of your domain, for example **test.lab**, if the box above was not checked.
- Click **Apply**.

2. On the same page under **IPv4 Settings**:

- Select **Enable IPv4**.
- **DHCP enabled** - (optional, depending on your network configuration). This box is checked for the Dell test environment.
- **Use DHCP to obtain DNS server address** - only check this box if you are using a DHCP server *and* it is configured to point to the Active Directory Server running DNS. This is not checked for the Dell test environment.
- **Static Preferred DNS Server** - Specify the IP address of your domain controller running DNS if the box above was left unchecked. For the Dell test environment, it is 172.23.199.28.
- **Alternate DNS server** - optional, can leave at 0.0.0.0.
- Click **Apply**.

This is how the iDRAC Network Settings look for the Dell test environment set up:

Figure 8. iDRAC network settings.

System
PowerEdge R720
root , Admin

Network

Common Settings

Attribute	Value
Register DRAC on DNS	☐
DNS DRAC Name	idrac-JCS0201
Auto Config Domain Name	☐
Static DNS Domain Name	test.lab

IPv4 Settings

Attribute	Value
Enable IPv4	☒
DHCP Enable	☒
Static IP Address	192.168.0.120
Static Gateway	192.168.0.1
Static Subnet Mask	255.255.255.0
Use DHCP to obtain DNS server addresses	☐
Static Preferred DNS Server	172.23.199.28
Static Alternate DNS Server	0.0.0.0

IPv6 Settings

Attribute	Value
Enable IPv6	☐

Configure the iDRAC Directory Services settings

- Go to **iDRAC Settings > User Authentication > Directory Services** (Reminder that an Enterprise License is required to get the Directory Services option).
 - Select **Microsoft Active Directory**, and then click **Apply**.
 - In the **Active Directory Configuration and Management** page, scroll down to bottom of page and click **Configure Active Directory**.
 - Select **Enable Certificate Validation**.
 - Upload the **Directory Service CA Certificate** - For this, you upload the certificate file generated earlier (named ad2.cer in this example) to iDRAC. First, copy this file from the Domain Controller to your management station. Second, in the iDRAC Web GUI next

to **Upload Directory Service CA Certificate**, click **Browse**, point to the file, and click **Upload**.

You should get a message similar to the following:

Figure 9. Upload complete.



If you get a message indicating the Certificate is not valid, there may be a date/time discrepancy between your CA and the iDRAC. Make sure the date and time on the iDRAC match the date and time on the CA (the Domain Controller in this document) and try again.

Note: If the certificate was issued from a *newly created* CA, it may continue to be reported as not valid even though the iDRAC and CA server dates and times match. This is because the iDRAC treats its time as UTC (Coordinated Universal Time). For example, if your CA server was created today at 2:00 pm Central Standard Time, the iDRAC views this as 2:00 pm *UTC*, a difference of 6 hours. As a result, the "valid from" timestamp on the certificate is not considered valid by the iDRAC until 8:00 pm on the day the CA was created. You can work around this by temporarily moving the time on the Managed System containing the iDRAC ahead by the appropriate amount for your time zone and resetting the iDRAC, or by simply waiting until the time has passed. Dell is aware of this issue and is developing a fix for a later release.

2. After you receive the **Upload complete** message pictured above, click **OK**.
3. Click **Next**.
4. Select **Enable Active Directory**.
5. Clear **Enable Single Sign-on**.
6. **User Domain Name**. Click Add and enter the FQDN of your domain. For example, **test.lab**, and click **OK**.
7. Select **Specify Domain Controller Addresses** and enter the FQDN of your Domain Controller for **Domain Controller Server Address 1** (For example, **ad2.test.lab**).
8. Click **Next**.
9. Select **Standard Schema**.
10. Click **Next**.
11. Select **Specify Global Catalog Server Addresses** and enter the **FQDN of your Domain Controller** for **Global Catalog Server Address 1** (For example, **ad2.test.lab**).
12. Click **Role Group 1**.
 - For the **Group Name**, enter **iDRACAdministrators** (Note: all group names must be an exact match to the group names you created earlier in Active Directory).

- **Group Domain** - enter your domain name. For example, **test.lab**.
- **Role Group Privilege Level** - Select **Administrator** from the drop down box. Note that all nine of the privilege check boxes are checked. Even though these privileges can be customized, it is recommended that you keep the default boxes checked for the Administrator and Read Only users. "Operator" is the correct user level to make customized privilege selections.
- Click **Apply**.

13. Click **Role Group 2**.

- **Group Name** - iDRACOperators
- **Group Domain** - test.lab for example
- **Privilege Level** - Select **Operator** from the drop-down list. Note that seven of the nine privilege boxes are checked. This is where customized privileges (if any) should be made by checking/unchecking the appropriate boxes as desired.
- Click **Apply**.

14. Click **Role Group 3**.

- **Group Name** - iDRACReadOnlyUsers
- **Group Domain** - test.lab for example
- **Privilege Level** - Select **Read Only** from the drop-down list.
- Click **Apply**
- Click **Finish**. You are taken to a summary page that looks similar to this:

Figure 10. Directory Services Summary.

The screenshot shows the iDRAC7 Directory Services Summary page. The browser address bar displays the URL: `https://172.26.9.56/index.html?ST1=2ff9c1840a874a6102786f625b3d72d6`. The page header includes the Dell logo and 'INTEGRATED DELL REMOTE ACCESS CONTROLLER 7 Enterprise'. The left sidebar shows a navigation menu with options like Overview, Server, Power / Thermal, Alerts, Setup, Troubleshooting, Licenses, Intrusion, iDRAC Settings, Network, User Authentication, iDRAC Firmware Update, Sessions, Hardware, Batteries, Fans, CPU, Front Panel, Network Devices, Power Supplies, Removable Flash Media, Storage, Physical Disks, Virtual Disks, Controllers, and Enclosures. The main content area is titled 'Directory Services' and contains several sections:

- Common Settings:** A table with attributes and values.

Attribute	Value
Active Directory Enabled	Yes
Single Sign-On Enabled	No
Schema Selection	Standard Schema
User Domain Name	test.lab
Timeout	120
Domain Controller Server Address 1 (FQDN or IP)	ad2.test.lab
Domain Controller Server Address 2 (FQDN or IP)	
Domain Controller Server Address 3 (FQDN or IP)	
Certificate Validation Enabled	Yes
- Active Directory CA Certificate:** A section showing certificate details.

Attribute	Value
Serial Number	64D72E3ECB017CB541124EE59B897B4E
Subject Information:	
Common Name (CN)	test-AD2-CA
Issuer Information:	
Common Name (CN)	test-AD2-CA
Valid From	Nov 9 23:04:22 2011 GMT
Valid To	Nov 9 23:14:20 2016 GMT
- Extended Schema Settings:** A table with attributes and values.

Attribute	Value
IDRAC Name	idrac1
IDRAC Domain Name	test.lab
- Standard Schema Settings:** A table with attributes and values.

Attribute	Value
Global Catalog Server Address 1 (FQDN or IP)	ad2.test.lab
Global Catalog Server Address 2 (FQDN or IP)	
Global Catalog Server Address 3 (FQDN or IP)	
- Standard Schema Role Groups:** A table with role groups, group names, group domains, and group privileges.

Role Groups	Group Name	Group Domain	Group Privilege
Role Group 1	IDRACAdministrators	test.lab	Administrator
Role Group 2	IDRACOperators	test.lab	Operator
Role Group 3	IDRACReadOnlyUsers	test.lab	Read Only
Role Group 4			None
Role Group 5			None

At the bottom right, there are buttons for 'Configure Active Directory' and 'Test Settings'.

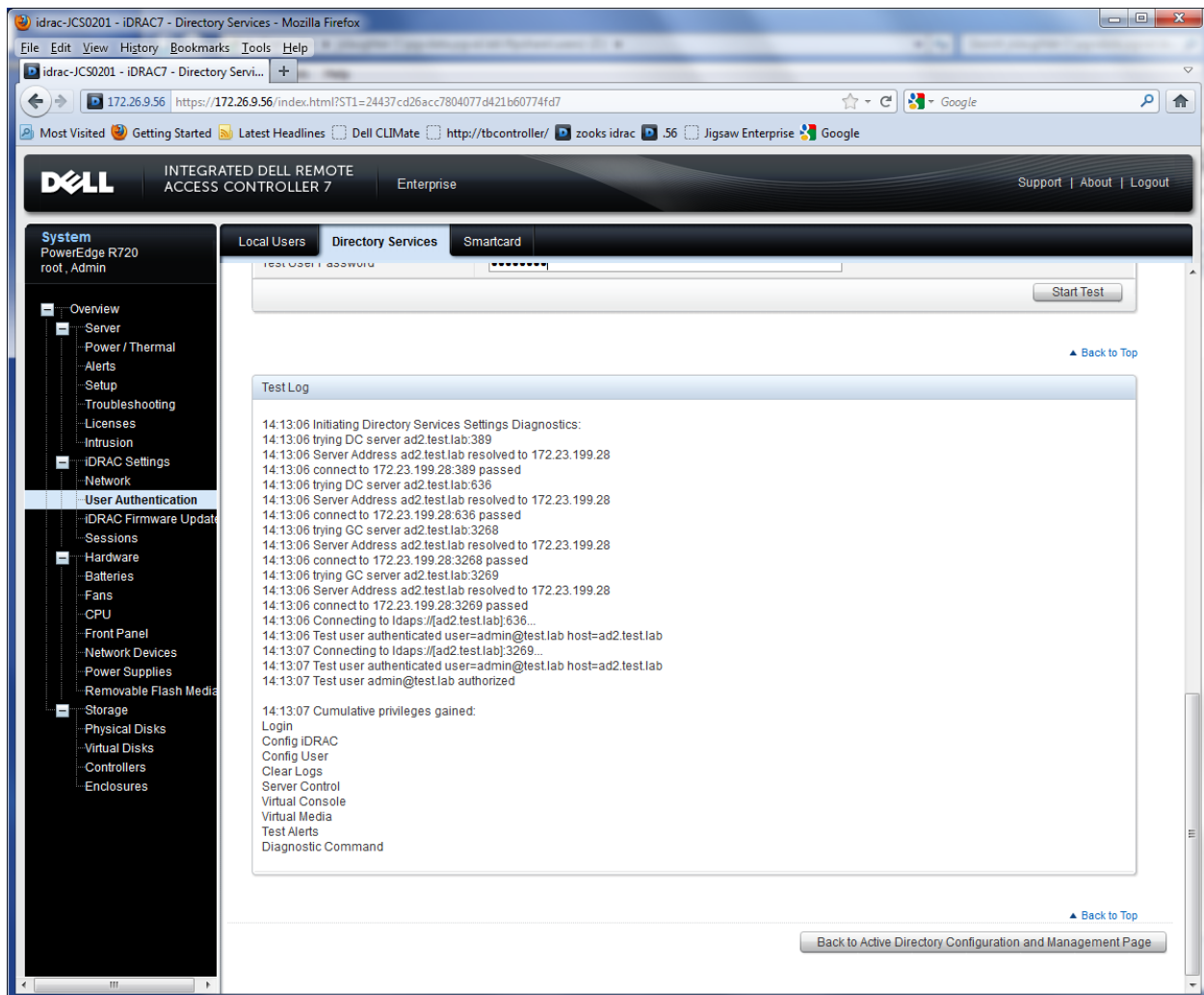
Test your Standard Schema configuration

1. Click the **Test Settings** button in the lower right part of the screen.
2. In the **Test User Name** text box, type your administrative user in username@domain.com format, for example, **admin@test.lab**.
3. In the **Test User Password** text box, type the user's password for the domain.
4. Click **Start Test**.
5. At the top of the results page, all tests must pass (including Certificate Validation) or be marked Not Applicable/Not Configured.

Integrating iDRAC7 with Active Directory

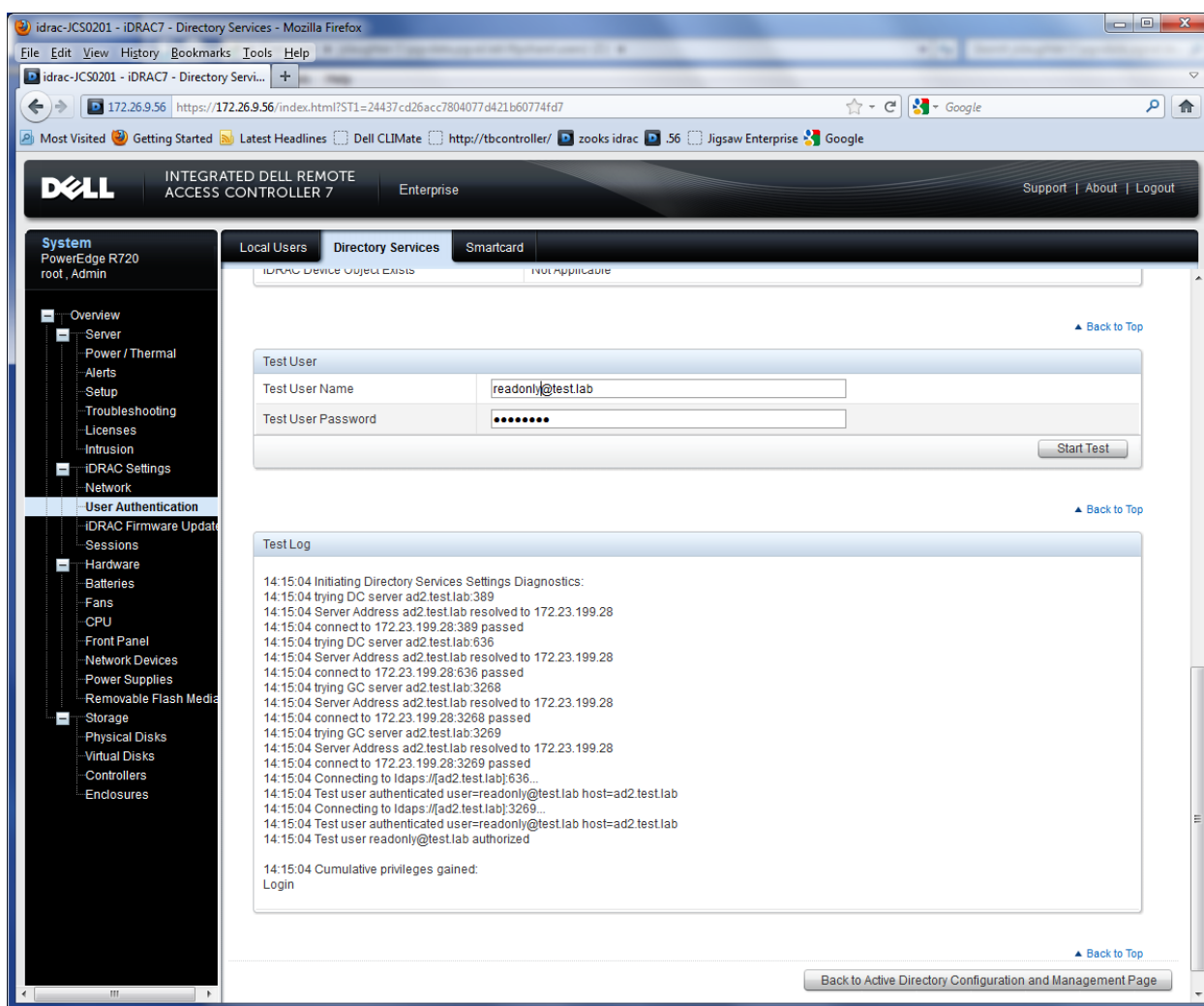
- The Test Log at bottom of page should have no errors and list all 9 privileges in the **Cumulative privileges gained** section as shown below.

Figure 11. Administrative User Test Results.



You can repeat the test with the other users you've created. The screenshot below shows the result from the read-only user. Note that the only privilege listed is **Login** which is the correct behavior for this user.

Figure 12. Read-Only User Test Results.



Active Directory login syntax options

Now that you have Active Directory authentication functional with your iDRAC, you need to know the different methods for authenticating as an active directory user. All of the iDRAC interfaces (GUI, racadm, WSMAN, SSH, and Telnet) accept the following domain-username formats:

Table 1. Domain username formats

Format	Example
username@domain.com	admin@test.lab
domain.com/username	test.lab/admin
domain.com\username	test.lab\admin

Note that the domain name must be fully qualified. For example, *test/admin* does not work; it must be *test.lab/admin*.

The login syntax is the same regardless of the schema option (standard or extended) used.

Authentication examples

For the following examples, the IP address of the iDRAC is 172.26.9.56. The domain is test.lab, and the user is admin with a password of Dell1234.

Example Racadm Command using Active Directory credentials

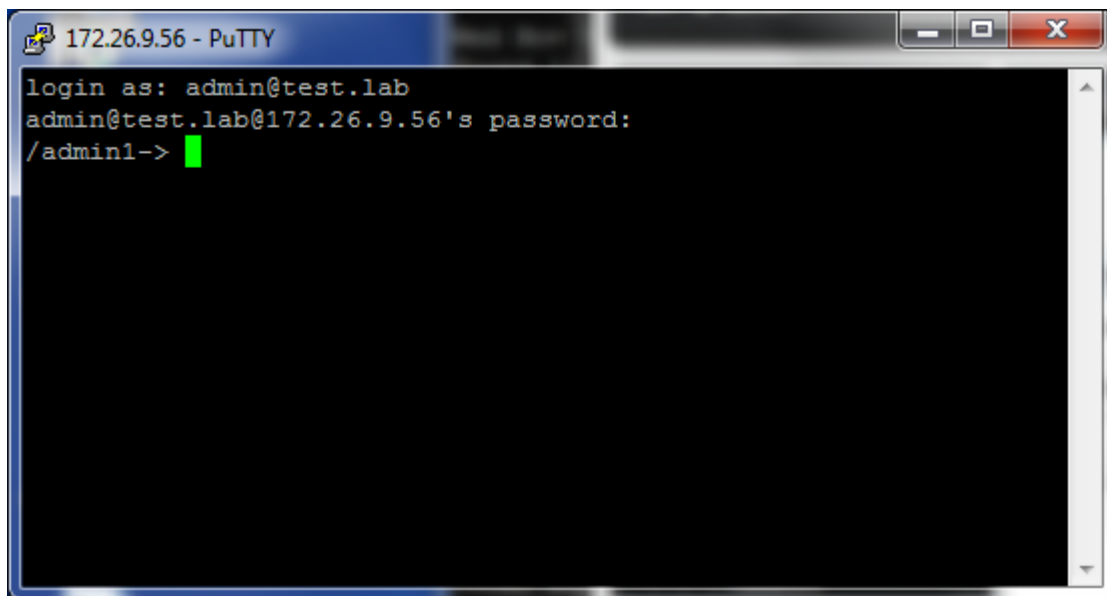
```
racadm -r 172.26.9.56 -u test.lab\admin -p Dell1234 getsysinfo
```

Example WSMAN (WinRM) Command using Active Directory credentials

```
winrm e cimv2/root/dcim/DCIM_PhysicalDiskView -u:test.lab/admin -p:Dell1234 -  
r:https://172.26.9.56/wsman -SkipCNcheck -SkipCAcheck -encoding:utf-8 -  
a:basic
```

Example SSH login using Active Directory credentials

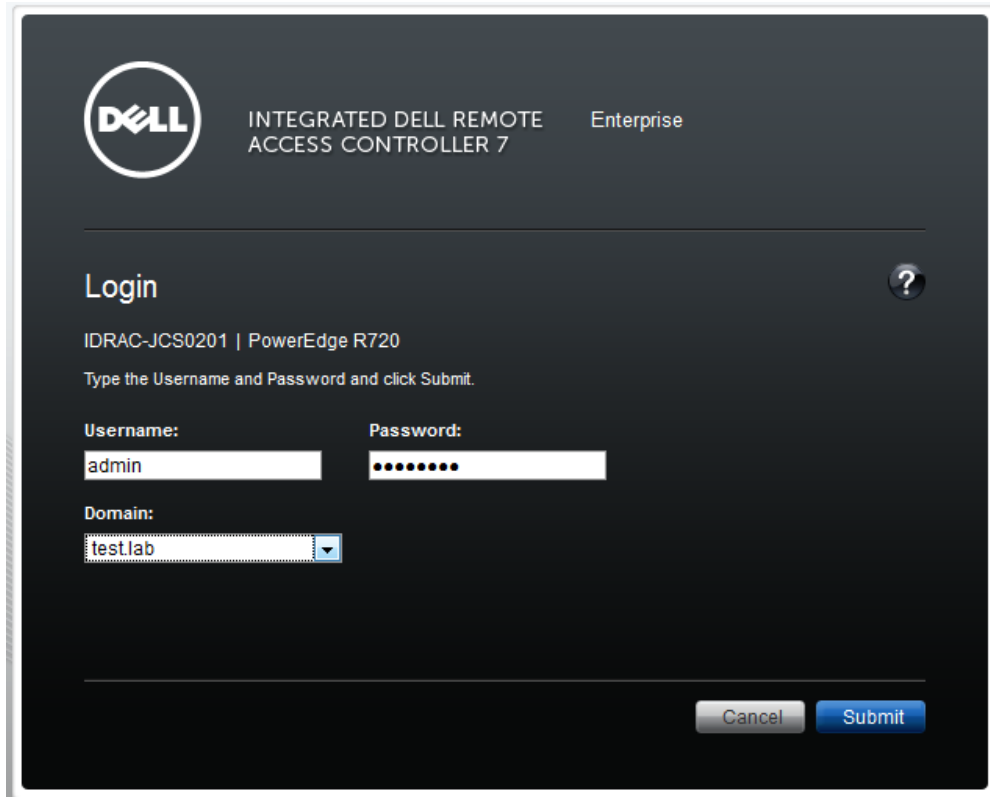
Figure 13. SSH login



iDRAC GUI login examples using Active Directory credentials

There is one additional option when logging into the iDRAC GUI. You can select the domain from the drop-down list and enter only the username and password in the boxes provided as shown below:

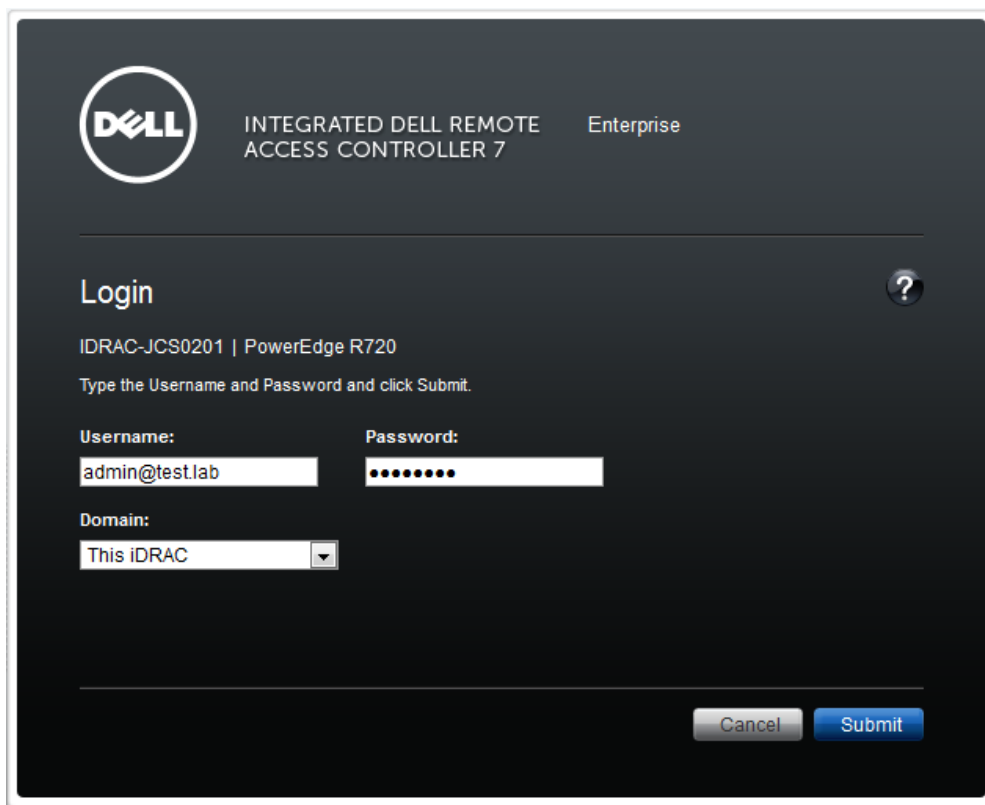
Figure 14. iDRAC GUI login option 1.



The screenshot displays the iDRAC GUI login interface. At the top left is the Dell logo. To its right, the text reads "INTEGRATED DELL REMOTE ACCESS CONTROLLER 7" and "Enterprise". Below this, the "Login" section is titled. Under the title, it says "iDRAC-JCS0201 | PowerEdge R720" and "Type the Username and Password and click Submit." There are three input fields: "Username:" with the value "admin", "Password:" with masked characters, and "Domain:" with a dropdown menu showing "test.lab". At the bottom right, there are "Cancel" and "Submit" buttons.

Or you can use one of the formats provided for the user name as long as you leave the Domain set to *This iDRAC* as follows:

Figure 15. iDRAC GUI login option 2.

The image shows the iDRAC GUI login screen. At the top left is the Dell logo. To its right, the text reads "INTEGRATED DELL REMOTE ACCESS CONTROLLER 7" and "Enterprise". Below this is a "Login" section with a help icon (question mark in a circle). Under "Login", it says "IDRAC-JCS0201 | PowerEdge R720" and "Type the Username and Password and click Submit." There are three input fields: "Username:" with the value "admin@test.lab", "Password:" with masked characters, and "Domain:" with a dropdown menu showing "This iDRAC". At the bottom right are "Cancel" and "Submit" buttons.

Configuring the domain controller for use with Active Directory Extended Schema

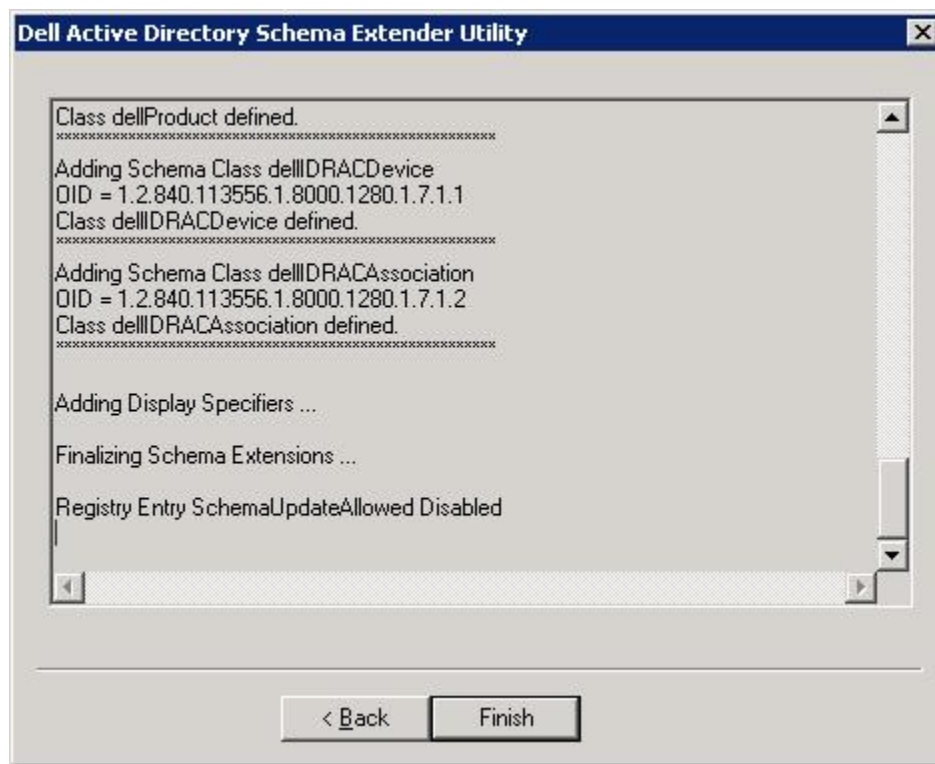
This section builds on the standard schema setup illustrated above. It uses the users, groups, certificates, and some of the iDRAC settings made above. Keep in mind that schema extensions cannot be undone. If you are using a virtual server it is a good idea to take a snapshot of the image before proceeding.

Extending the schema

1. Obtain the **Dell Systems Management Tools and Documentation DVD**, version 7.0.0 or later, provided with your Dell PowerEdge system.
2. Log in to your **Domain Controller** as an Administrator.
3. Run the **Schema Extender** from the DVD:
DVD_DRIVE:\SYSMGMT\ManagementStation\support\OMActiveDirectory_Tools\Remote_Management_Advanced\Schema Extender.exe
4. If a Security Warning message is displayed, click **Run**.
5. A Welcome Message is displayed, click **Next**.
6. A Warning Message is displayed indicating Schema Extensions cannot be undone, click **Next**.
7. Accept the default option to use current credentials, and then click **Next**.

8. The schema will be extended and you get a message similar to the following when complete:

Figure 16. Schema Extension Complete



9. Click Finish.

Optional - View the changes made to the Active Directory Schema

If you would like to see the changes made by extending the schema, you need to install the Microsoft Active Directory Schema snap in. You can do so as follows:

1. At the **command prompt**, type the following command, and then press **ENTER**:

```
regsvr32 schmmgmt.dll
```

2. A message indicates that the command succeeded. Click **OK**.
3. Open your saved **Console1.msc** (or create a new one by running mmc).
4. Click **File > Add/Remove Snap In**.
5. Select **Active Directory Schema**, click **Add**, then **OK**.

You can now view the changes as follows:

6. Expand **Active Directory Schema** and expand **Classes**. In the right pane, you can locate the added classes prefixed with "dell".

7. Under Active Directory Schema, click the **Attributes** folder. In the right pane, you can locate the added attributes prefixed with "dell".

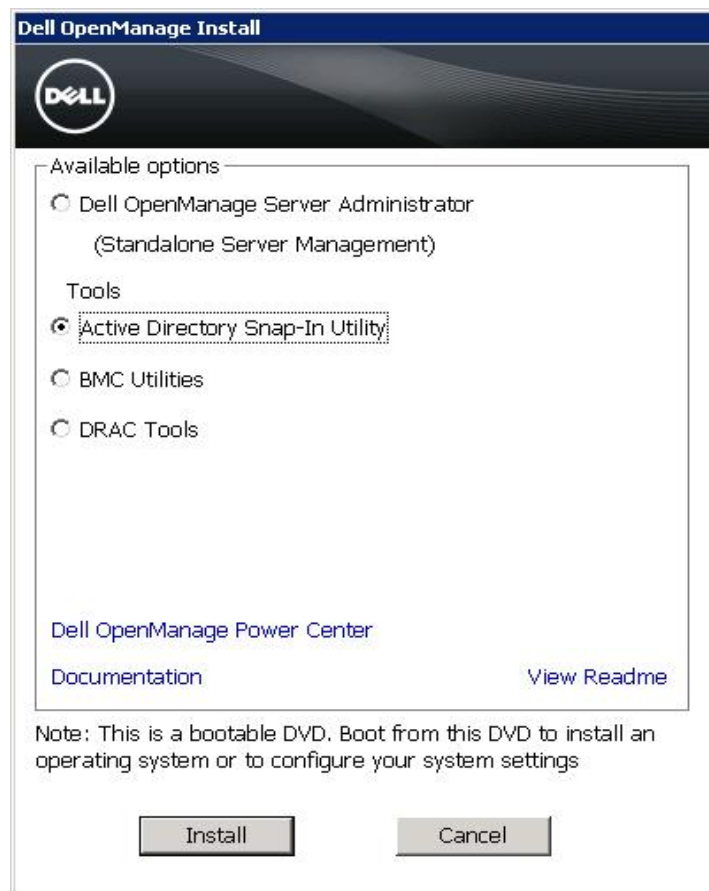
Install the Dell Extension to the Active Directory Users and Computers Snap-In for 32-bit Windows

Note: For Domain Controllers running 64-bit windows see the next section.

If your Domain Controller is running a 32-bit version of Windows, install the Dell extension to the Active Directory Users and Computers Snap-In as follows:

1. Run **Autorun.exe** from the root of the Systems Management Tools and Documentation DVD.
2. If a Security Warning message is displayed, click Run.
3. Select **Install Active Directory Snap-In Utility**, and then click **Install**.

Figure 17. Installing the 32-bit version.



Note: If you are using a Remote Desktop connection to connect to your Domain Controller and get an error that installation is not permitted from Remote Desktop, map a drive letter to the DVD rather than using a UNC share name and try again.

4. Click **Next**.

5. Accept the License agreement, and then click **Next**.
6. Click **Install**.
7. A success message is displayed when complete. Click **Finish**.

Install the Dell Extension to the Active Directory Users and Computers Snap-In for 64-bit Windows

Note: For Domain Controllers running 32-bit Windows see the preceding section.

If your Domain Controller is running a 64-bit version of Windows, install the Dell extension to the Active Directory Users and Computers Snap-In as follows.

1. Obtain the Systems Management Tools and Documentation DVD and run:

DVD_DRIVE:\SYSMGMT\ManagementStation\support\OMActiveDirectory_SnapIn64\support\vcredist.exe (This is the Visual C++ redistributable package).

DVD_DRIVE:\SYSMGMT\ManagementStation\support\OMActiveDirectory_SnapIn64\sample_install_activedirectory_snapins_64bit.bat

Install the Active Directory Users and Computers Snap-In to MMC

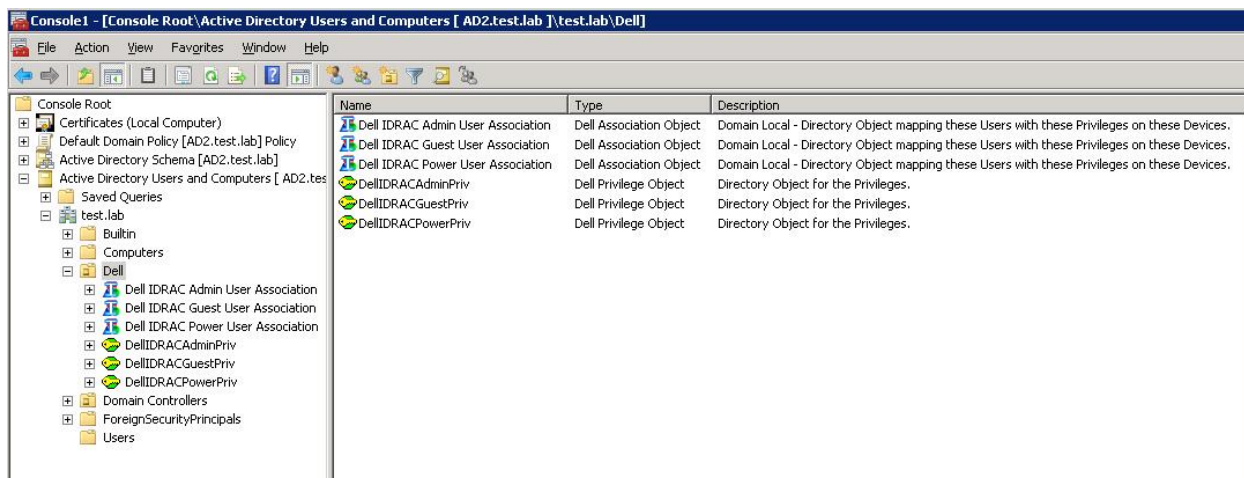
Note: The Dell Active Directory Snap-in extension is not fully functional in the Server Manager console. For full functionality use Microsoft Management Console per the following steps.

Use your saved **Console1.msc** file or create a new console by running **mmc**.

Add the **Active Directory Users and Computers** Snap-In to your console as follows:

1. Click **File > Add/Remove Snap In**.
2. Select **Active Directory Users and Computers**, click **Add**, then **OK**.
3. Expand **Active Directory Users and Computers** > expand the domain name (**test.lab**). Note that there is a new container named Dell containing six iDRAC objects. There are three Association objects (Admin, Guest, and Power User) and three levels of corresponding Privilege objects, as shown below.

Figure 18. Dell iDRAC Objects.



Privilege and role name changes

The privilege names and role names are a little different in Active Directory versus the iDRAC GUI as they have been renamed for iDRAC7. The old names have been retained in the Active Directory schema extension for backwards compatibility.

The following tables map the prior generation privilege names and role names to the current generation.

Table 2. Role Names.

Prior PowerEdge Generations (Seen in Active Directory)	Current PowerEdge Generation (Seen in the iDRAC GUI)
Admin	Administrator
Power User	Operator
Guest	Read Only

Table 3. Privilege names.

Prior PowerEdge Generations (Seen in Active Directory)	Current PowerEdge Generation (Local User Privileges seen in the iDRAC GUI)
Login	Login
Configure Device	Configure
Configure Users	Configure Users
Clear Logs	Logs
Execute Server Control Commands	System Control
Access Console Redirection	Access Virtual Console
Access Virtual Media	Access Virtual Media
Test Alerts	System Operations
Execute Diagnostic Commands	Debug

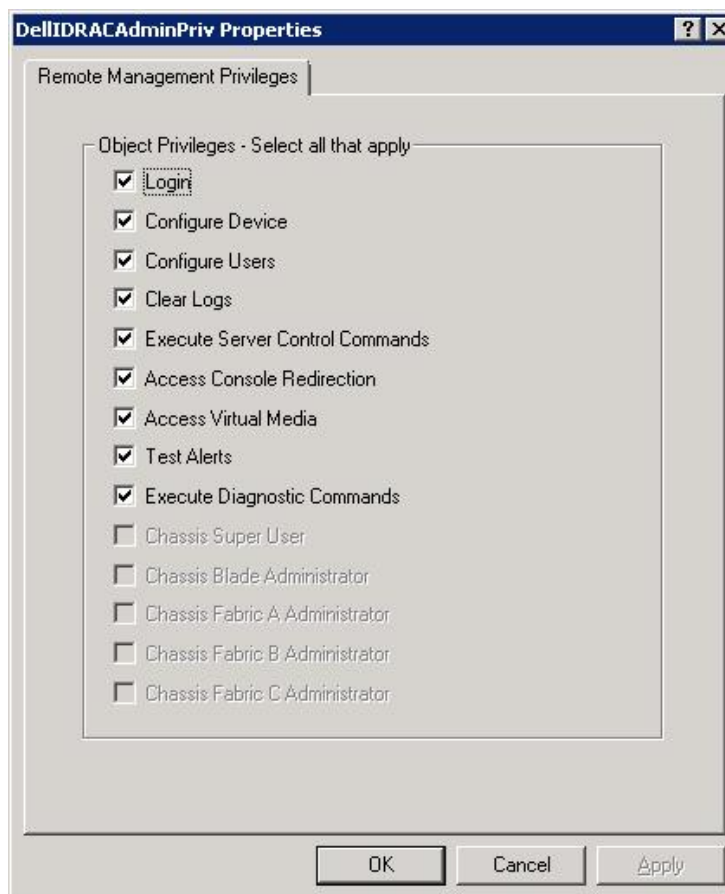
Active Directory objects

Privilege Objects

Right click on **DellIDRACAdminPriv** and select **Properties**.

You will see the following:

Figure 19. Dell Admin Privilege Object.



The privilege object lists all privilege names. All boxes are checked in this example since this is the DellIDRACAdminPriv, which controls the Administrator's privileges.

If you repeat the above with the DellIDRACGuestPriv, you will see only the Login box has been checked. Likewise, DellIDRACPowerPriv has by default all but two boxes checked.

If you want to customize user privileges it is recommended that you do so in the DellIDRACPowerPriv object by checking/unchecking the appropriate boxes. This represents the mid-level Power User (also known as the Operator) iDRAC role.

iDRAC objects

An iDRAC object is created below for each physical iDRAC that is integrated with Active Directory.

Association objects

An Association object is used to link iDRAC objects to Active Directory users (or groups) to Privilege objects. The Association object effectively bundles the three items together.

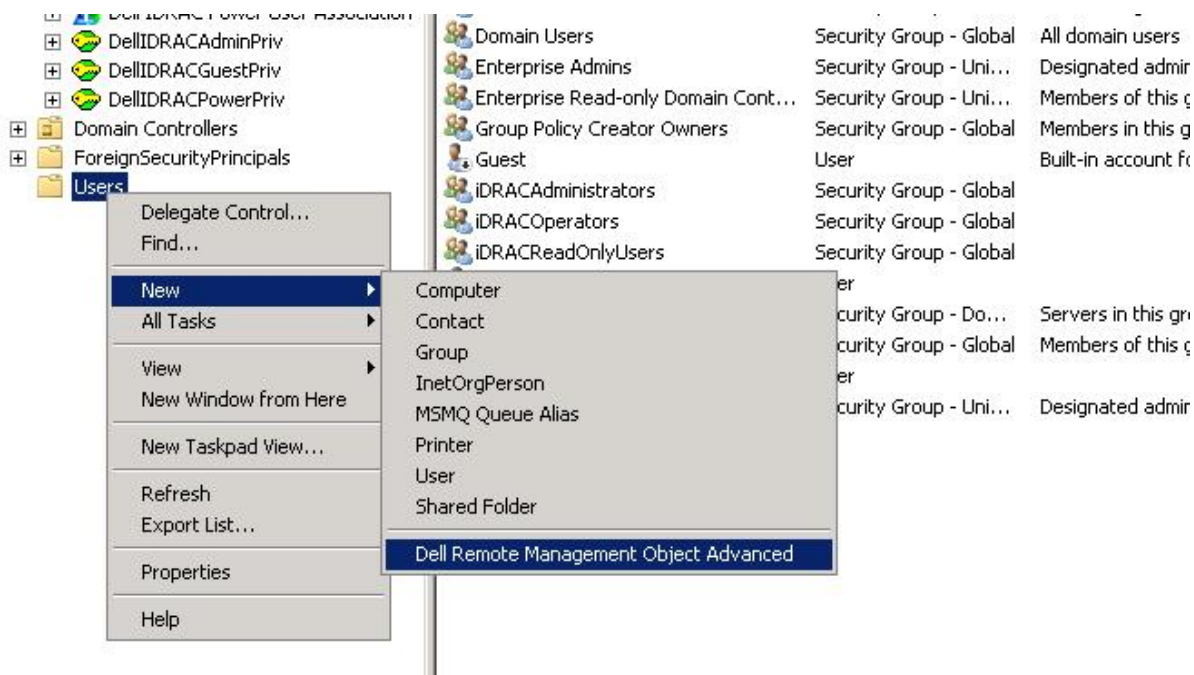
Configure Active Directory

In the following steps, an iDRAC object representing the physical iDRAC in the Managed PowerEdge server is created. Using the Admin User Association object, the iDRAC object is associated with the iDRACAdministrators group and the DellIDRACAdminPriv object. These steps are repeated for other groups requiring lesser privileges for the same iDRAC object.

1. Right Click on the **Users** Container and select **New > Dell Remote Management Object Advanced**.

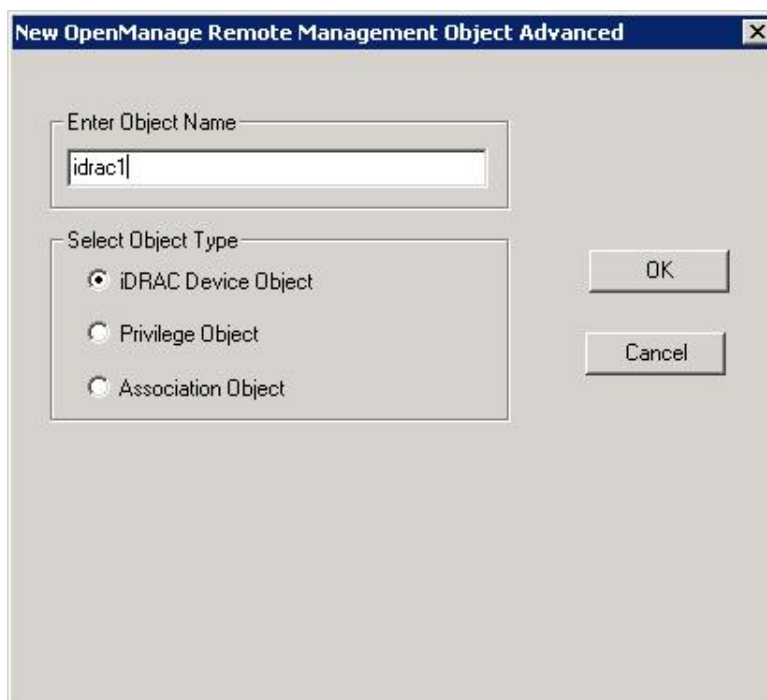
Reminder - this option is not available if you are using the Server Manager console - be sure you are using MMC.

Figure 20. Adding a New Dell Object.



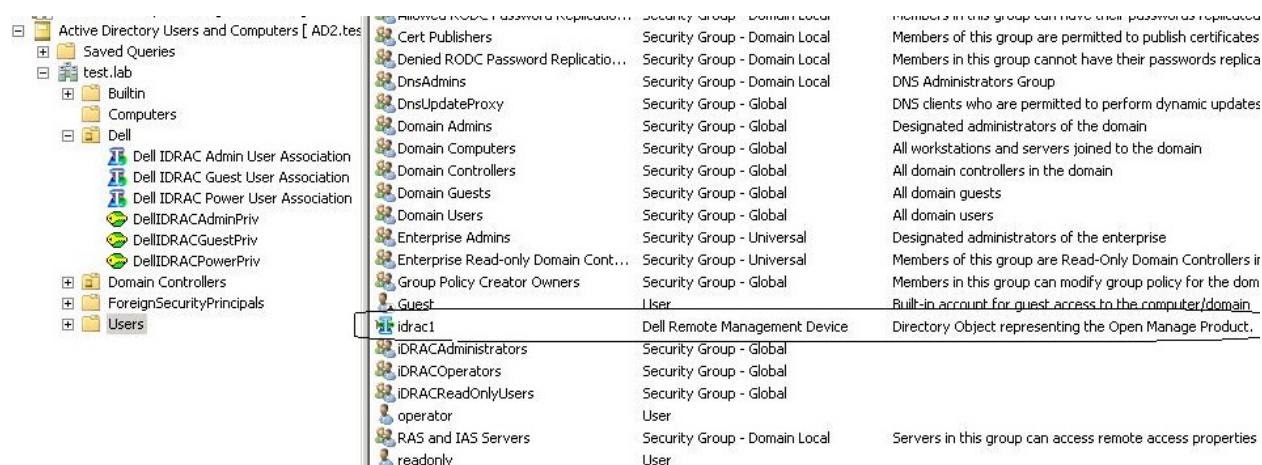
2. For **Enter Object Name**, put the name to be used for the iDRAC. This name must be unique for each iDRAC object in Active Directory. For example, use **idrac1**.

Figure 21. Entering Object Name



3. Make sure the **iDRAC Device Object** option button is selected and click **OK**. The iDRAC device object now appears in the Users container in Active Directory.

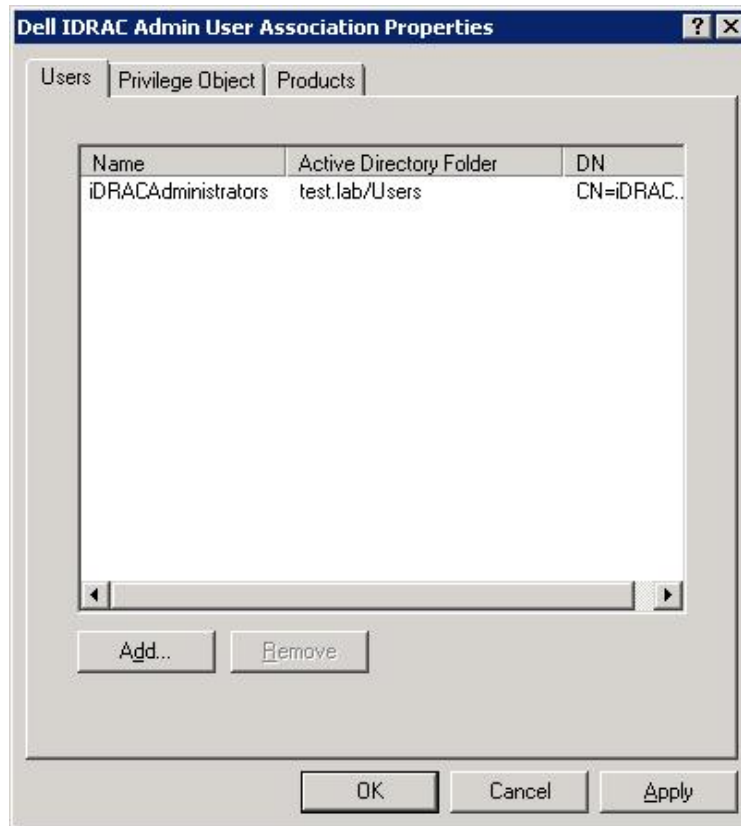
Figure 22. iDRAC Device Object.



4. Expand the Dell Container under yourdomain.com (**test.lab**), right click on **Dell iDRAC Admin User Association** and select **Properties**.
 - On the **Users** tab, click **Add > Object Types > select Groups**, and then click **OK**.
 - Under **Enter the object names to select** enter **iDRACAdministrators**, click **Check Names**.
 - The object should be found as shown by an underline, and then click **OK**.

Your window should look something like this:

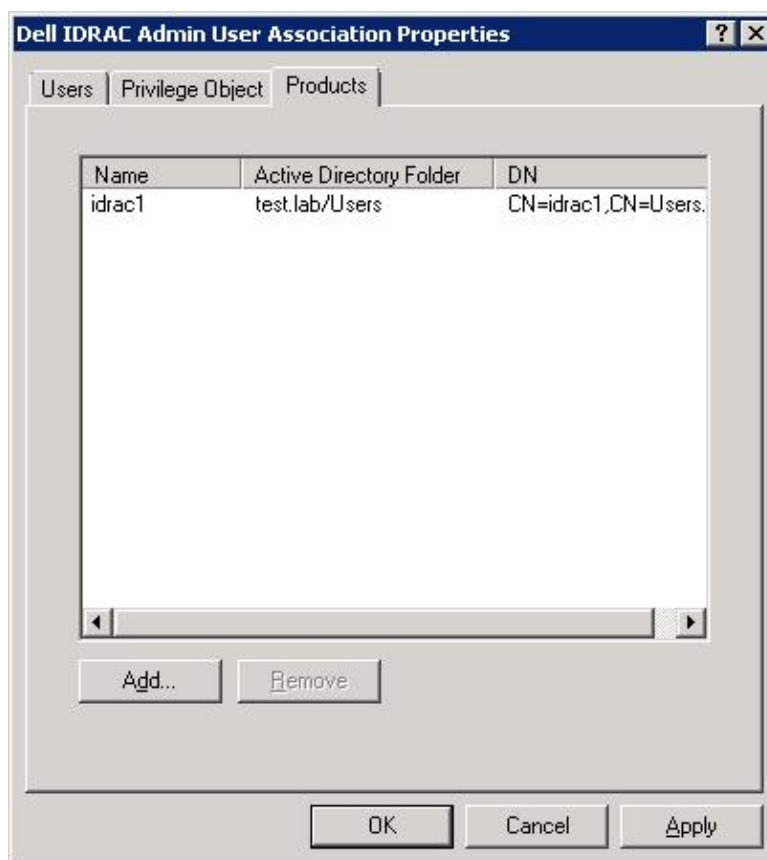
Figure 23. Configuring the Admin User Association Object.



5. Leave the window open and click the **Privilege Object** tab. It will be pre-populated with the **DellAdminPriv** object.
6. On the **Products** tab, select **Add**.
7. Under **Enter the object names to select**, enter the iDRAC object name used earlier (**idrac1**) and then click **Check Names** (it should be found as shown by an underline), click **OK**.

Your window should look something like this:

Figure 24. Configuring the iDRAC Admin User Association Object (continued)



8. Repeat for the Guest User (also known as the Read Only User)
 - Right click on **Dell iDRAC Guest User Association** and select **Properties**.
 - On the Users tab, click **Add > Object Types > select Groups**, and then click **OK**.
 - In the **Enter the object names to select** box enter **iDRACReadOnlyUsers > Check Names** (it should be found) >click **OK**.
 - The Privilege Object tab will be pre-populated with the **DellGuestPriv object**.
 - On the Products tab, click **Add**.
 - Enter your **iDRAC name(idrac1) > Check Names** (it should be found) > **OK > OK**.
9. Repeat for the Power User (also known as the Operator)
 - Right click on **Dell iDRAC Power User Association** and select **Properties**.
 - On the Users tab, click **Add > Object Types > select the Groups box > OK > in the Enter the object names to select** text box enter **iDRACOperators > Check Names** (it should be found) > **OK**.
 - The **Privilege Object** tab will be pre-populated with the **DellPowerPriv object**.
 - On the **Products** tab, click **Add**.
 - Enter your iDRAC name (**idrac1**) > **Check Names** (it should be found) > **OK > OK**.

Adding users

You can add new users to the appropriate Active Directory group (iDRACAdministrators, and so on) with no further configuration necessary.

Adding iDRACs

If you have additional iDRACs you wish to set up, create a new iDRAC object with a unique name for each (such as idrac2, idrac3, and so on). You can then follow the steps above to add the additional iDRAC objects to the Products tab in each of the three Association objects. You can add multiple iDRACs at the same time by separating their names with semicolons in the **Enter the object names to select** box or by typing the first few letters in their names (assuming they all start with the same few letters), clicking **Check Names**, and selecting the iDRAC objects from the **Multiple Names Found** box.

Configuring iDRAC for use with Active Directory Extended Schema

In your browser at the management station, log into the iDRAC GUI of your managed system.

`https://<idrac_ip_address>`

The iDRAC Network Settings can remain as previously configured for Standard Schema authentication. If you would like to review them please see Configure the iDRAC Network Settings in the Standard Schema configuration section [here](#).

1. Go to **iDRAC Settings > User Authentication > Directory Services**.

- Make sure **Microsoft Active Directory** is checked and click the **link** or **Apply**.
- Scroll Down to bottom of page and click **Configure Active Directory**.
- Make sure **Enable Certificate Validation** is checked and the certificate uploaded for during Standard Schema configuration is shown under **Current Directory Service CA Certificate**.
- Click **Next**.
- Make sure **Enable Active Directory** is checked.
- Make sure **Enable Single Sign-On** is unchecked.
- For **User Domain Name**, make sure the **FQDN of your domain name** is present (for example, test.lab).
- Make sure **Specify Domain Controller Addresses** is selected and the **FQDN of your Domain controller** is still present for **Domain Controller Server Address 1**. (For Example, ad2.test.lab).
- Click **Next**.
- This is where the changes start. Select **Extended Schema**, and then click **Next**.
- For **iDRAC Name**, use the name of the iDRAC object you created in Active Directory (for example, **idrac1**).

- Specify your **iDRAC Domain Name** (For example, **test.lab**).
- Click **Finish**. You are taken to a summary page that looks similar to the following:

Figure 25. Active Directory Configuration and Management summary page.

The screenshot shows the Dell iDRAC7 web interface in a Mozilla Firefox browser. The address bar shows the URL: <https://172.26.9.56/index.html?ST1=3e978c93590641621036964b14a99213>. The page title is "idrac-JCS0201 - iDRAC7 - Directory Services - Mozilla Firefox".

The interface has a top navigation bar with "DELL", "INTEGRATED DELL REMOTE ACCESS CONTROLLER 7", and "Enterprise". Below this is a sidebar menu for "System" (PowerEdge R720) with options like Overview, Server, Power / Thermal, Alerts, Setup, Troubleshooting, Licenses, Intrusion, iDRAC Settings, Network, User Authentication, iDRAC Firmware Update, Sessions, Hardware, Batteries, Fans, CPU, Front Panel, Network Devices, Power Supplies, Removable Flash Media, Storage, Physical Disks, Virtual Disks, Controllers, and Enclosures.

The main content area is titled "Active Directory Configuration and Management" and has tabs for "Local Users", "Directory Services", and "Smartcard". The "Directory Services" tab is selected.

Common Settings

Attribute	Value
Active Directory Enabled	Yes
Single Sign-On Enabled	No
Schema Selection	Extended Schema
User Domain Name	test.lab
Timeout	120
Domain Controller Server Address 1 (FQDN or IP)	ad2.test.lab
Domain Controller Server Address 2 (FQDN or IP)	
Domain Controller Server Address 3 (FQDN or IP)	
Certificate Validation Enabled	Yes

Active Directory CA Certificate

Certificate	
Serial Number	: 64D72E3ECB017CBD41124EE59B897B4E
Subject Information:	
Common Name (CN)	: test-AD2-CA
Issuer Information:	
Common Name (CN)	: test-AD2-CA
Valid From	: Nov 9 23:04:22 2011 GMT
Valid To	: Nov 9 23:14:20 2016 GMT

Extended Schema Settings

Attribute	Value
iDRAC Name	idrac1
iDRAC Domain Name	test.lab

Note: At the bottom of the page (not visible above) you see sections labeled Standard Schema Settings and Standard Schema Role Groups. These are still retained in the iDRAC configuration but are not used when Extended Schema is selected. This lets you easily switch back and forth between the two schema options with minimal additional configuration.

Test your Extended Schema configuration

1. Scroll down to the bottom of the **Active Directory Configuration and Management** summary page, and click **Test Settings**.
2. In **Test User Name** text box, type your administrative user in username@domain.com format, for example, **admin@test.lab**.
3. In the **Test User Password text box**, type the user's password for the domain.
4. Click **Start Test**

At the top of the results page, all tests must Pass (including Certificate Validation) or be marked Not Applicable/Not Configured/Not Run.

The Test Log at bottom of page should have no errors and list all 9 privileges in the Cumulative privileges gained section as shown below.

Figure 26. Test Results for Administrative User.

Test User	
Test User Name	admin@test.lab
Test User Password	••••••••

Test Log
14:07:38 Initiating Directory Services Settings Diagnostics: 14:07:38 trying DC server ad2.test.lab:389 14:07:38 Server Address ad2.test.lab resolved to 172.23.199.28 14:07:38 connect to 172.23.199.28:389 passed 14:07:38 trying DC server ad2.test.lab:636 14:07:38 Server Address ad2.test.lab resolved to 172.23.199.28 14:07:38 connect to 172.23.199.28:636 passed 14:07:38 Connecting to ldaps://[ad2.test.lab]:636... 14:07:38 Test user authenticated user=admin@test.lab host=ad2.test.lab 14:07:38 Test user admin@test.lab authorized 14:07:38 Cumulative privileges gained: Login Config iDRAC Config User Clear Logs Server Control Virtual Console Virtual Media Test Alerts Diagnostic Command

It is recommended that you also run the test for your users with lower privilege levels (readonly and operator for example) at this time to confirm everything is configured correctly.

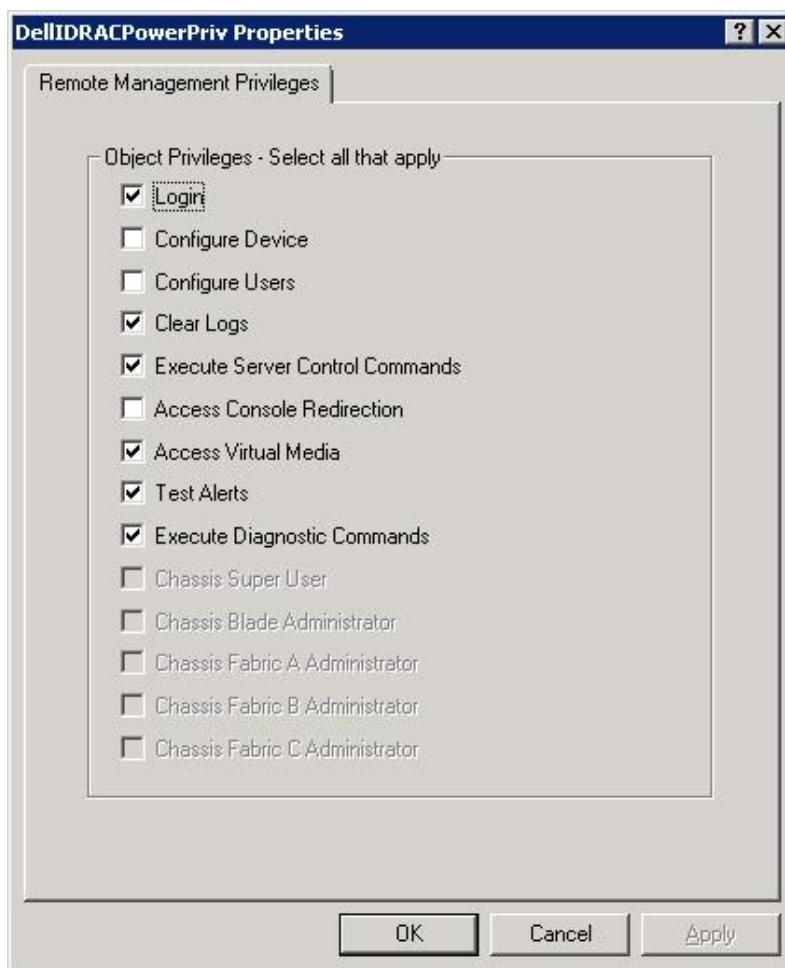
You can now authenticate to the iDRAC for all services (racadm, WSMAN, SSH, Telnet, and the GUI) as shown earlier in the [Authentication Examples](#) section.

Creating an Active Directory user with customized iDRAC privileges

This example creates a new user, assigns the user to the iDRACOperators group, modifies the default privileges held by the Power User role (also known as the Operator role), and then tests the configuration.

1. At the Domain Controller, under **Active Directory Users and Computers** create a new user with a login name of **John_Smith**. Assign a password and uncheck **User must change password at next logon**.
2. Add **John_Smith** to the **iDRACOperators** group.
3. Now customize the privileges John_Smith (and the iDRACOperators group) receives by removing the ability to use **Console Redirection**.
4. Under **Active Directory Users and Computers** in the **Dell** container, right-click on **DellIDRACPowerPriv** and select **Properties**.
5. Uncheck **Access Console Redirection**. The window should look like this:

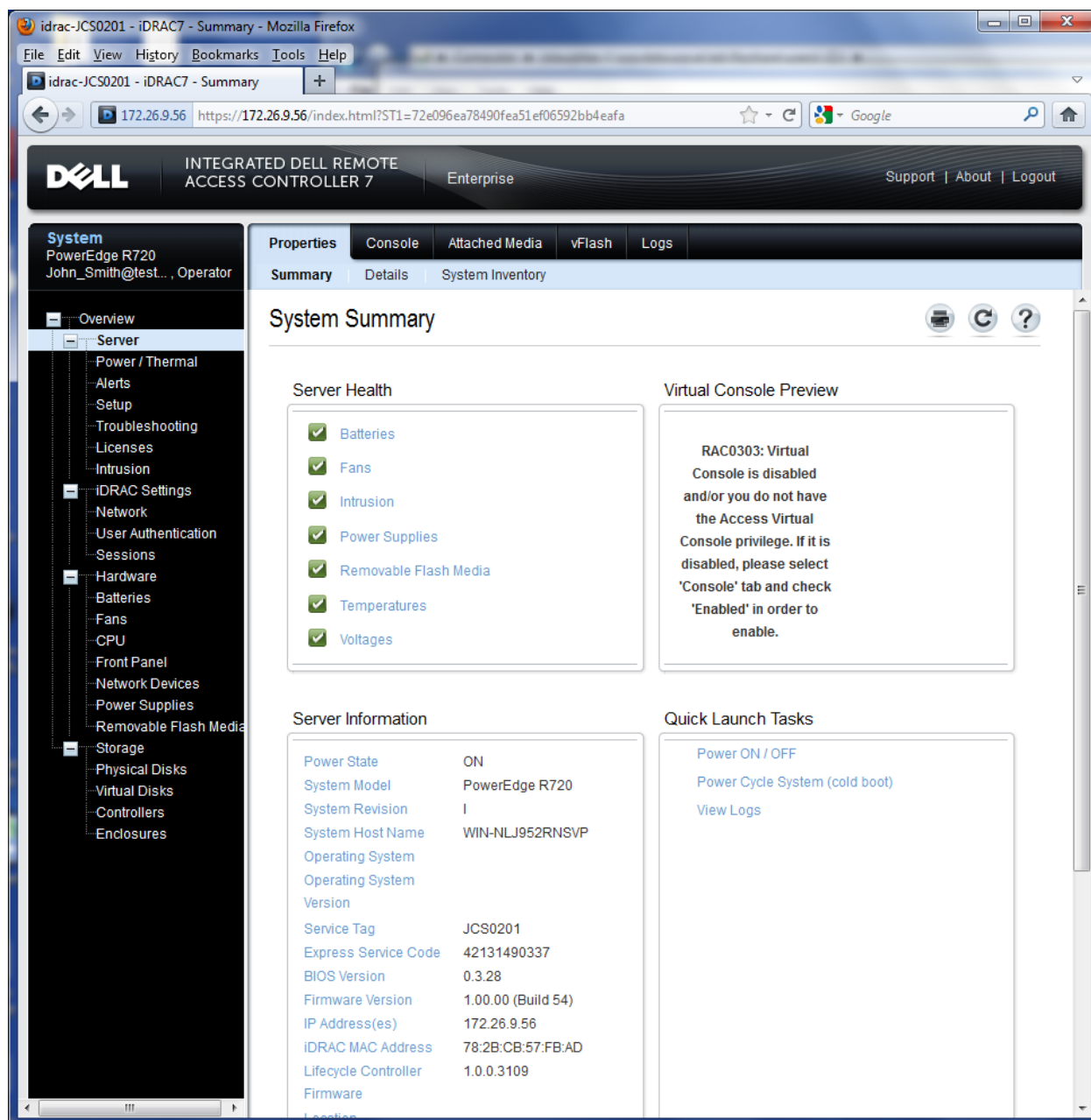
Figure 27. Configuring Custom Privileges.



Integrating iDRAC7 with Active Directory

6. To close, click **OK**.
7. At the **management station**, log out of the **iDRAC GUI**.
8. Log back in to the **iDRAC GUI** as John_Smith. Be sure to specify your domain (**test.lab**). You should see a System Summary Page similar to this:

Figure 28. John Smith's System Summary page

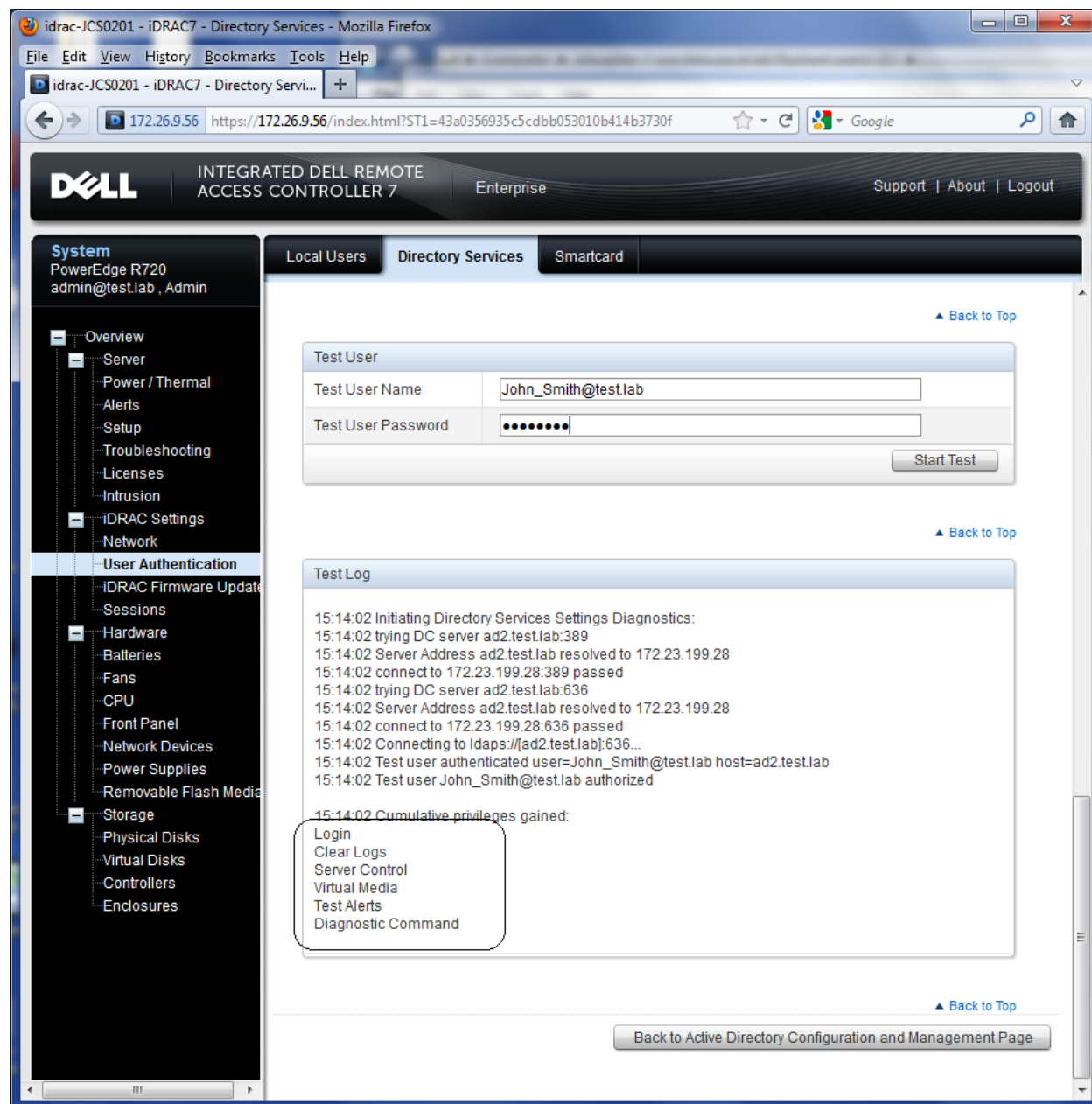


Note that the Virtual Console Preview (upper right of Summary Page) is not shown and was replaced with a message indicating the user does not have access. This feature is part of the *Access Console Redirection* privilege that was removed from the group that the user belongs to in the prior step.

9. Do a full Test of John_Smith's privileges to confirm everything is configured properly as follows:

- Log out and log back in as an *administrative* user in the iDRAC GUI. Go to **iDRAC Settings > User Authentication > Directory Services > Microsoft Active Directory > Test Settings**.
- Enter **John_Smith@yourdomain.com** (John_Smith@test.lab) and **John's password**, and then click **Start Test**.
- The results should match the privileges configured above and appear as follows:

Figure 29. Privilege Test Results for John Smith.



Summary

Active Directory integration with iDRAC7 can greatly simplify management of your iDRAC users and privileges. This document simplifies the set up process and enables you to evaluate the Standard Schema and Extended Schema options for use with the Dell iDRAC.